Summary in Graph

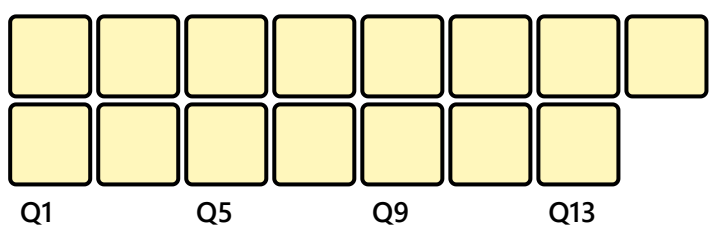
Exam Summary (GO Classes GATE CS | Theory of Computation | Finite Automata and Regular Expression | Weekly Quiz 3)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



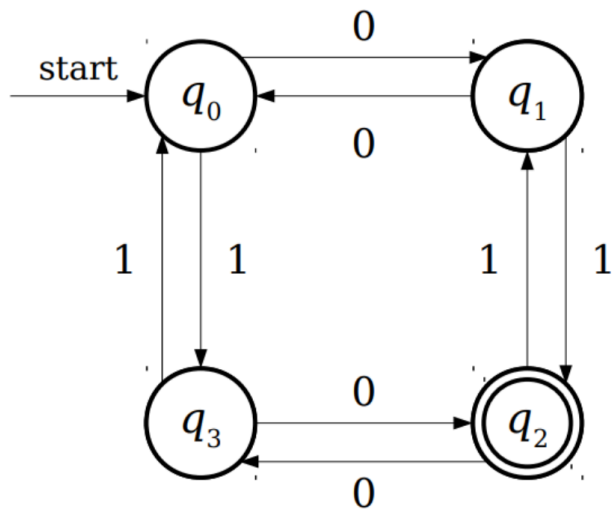
* Average Stats based on first-attempt stats of Top 10% (34) out of 337 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Select Type | Award: 1 | Penalty: 0 | Theory of Computation

Consider the following simple Finite Automaton.



Which of the following strings are not accepted by the given finite automaton?

A. 010110
B. 101000
C. 11011100
D. Empty String

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 Multiple Select Type | Award: 1 | Penalty: 0 | Theory of Computation

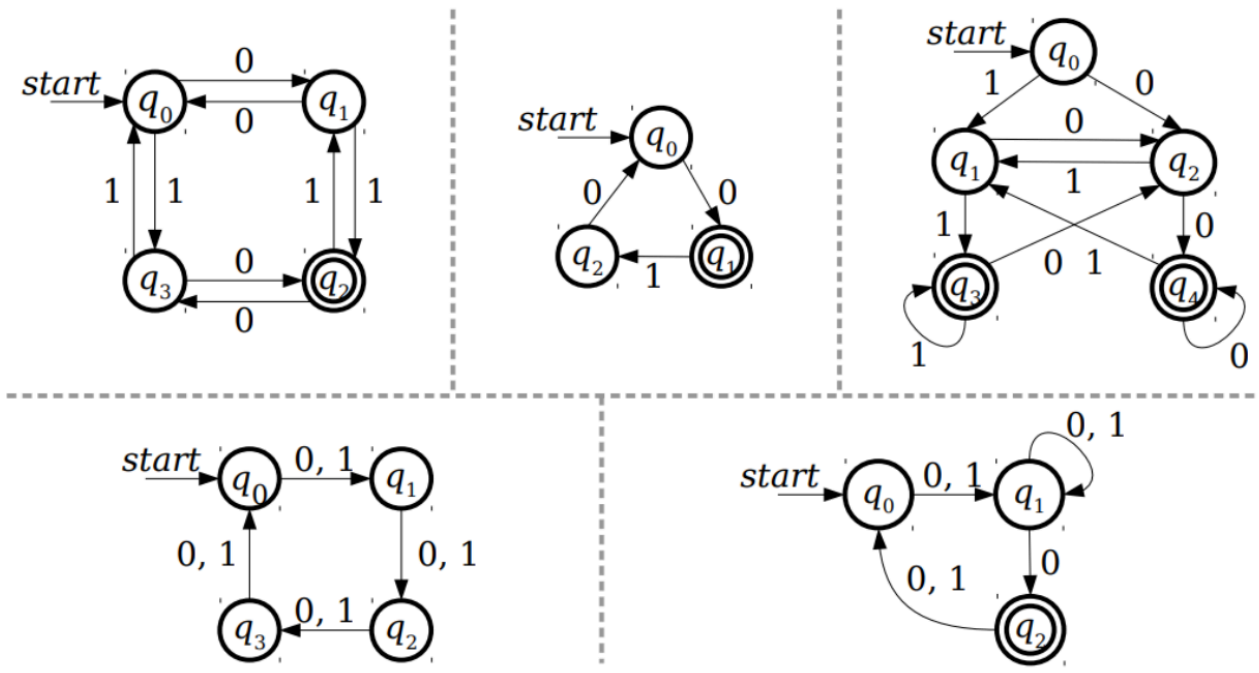
Which of the following statements is/are true?

A. A language of an automaton can have an infinitely long string (or many of them) in it.
B. A language of an automaton can contain infinitely many strings.
C. A language of an automaton can contain no strings.
D. A language of an automaton can contain finitely many strings.

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

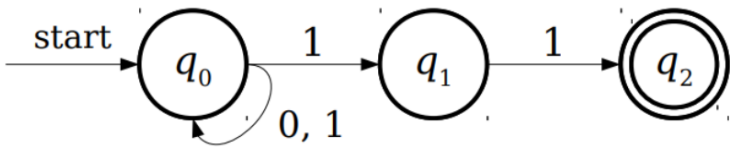
Q #3 Numerical Type | Award: 1 | Penalty: 0 | Theory of Computation

How many of these are DFAs over $\{0, 1\}$?



Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Multiple Select Type | Award: 1 | Penalty: 0 | Theory of Computation



As with DFAs, the language of an NFA N is the set of strings that N accepts:

$$L(N) = \{w \in \Sigma^* \mid N \text{ accepts } w\}.$$

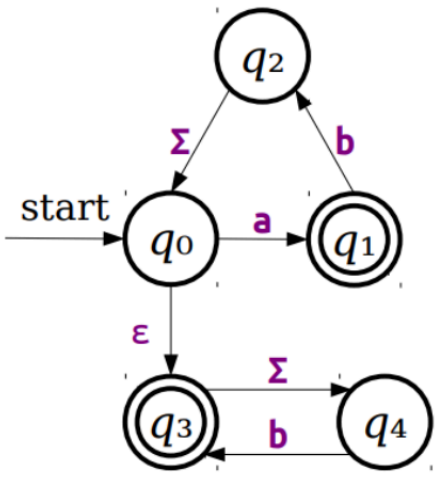
What is the language of the NFA shown above?

A. $\{01011\}$
B. $\{w \in \{0,1\}^* \mid w$ contains at least two 1s$\}$
C. $\{w \in \{0,1\}^* \mid w$ ends with $11\}$
D. $\{w \in \{0,1\}^* \mid w$ ends with $1\}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 Numerical Type | Award: 1 | Penalty: 0 | Theory of Computation

We transform the following NFA into an equivalent DFA using the subset construction (or the powerset construction) algorithm.

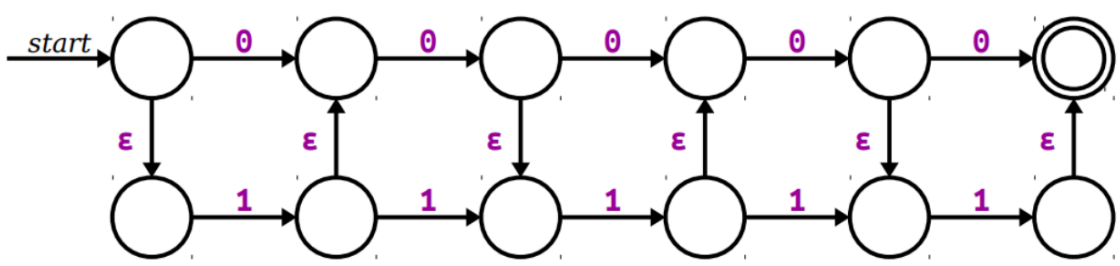


	a	b
$\{q_0, q_3\}$	$\{q_1, q_4\}$	$\{q_4\}$
$\{q_1, q_4\}$	Ø	$\{q_2, q_3\}$
$\{q_4\}$	Ø	$\{q_3\}$
$\{q_2, q_3\}$	$\{q_0, q_3, q_4\}$	$\{q_0, q_3, q_4\}$
$\{q_3\}$	$\{q_4\}$	$\{q_4\}$
$\{q_0, q_3, q_4\}$	$\{q_1, q_4\}$	$\{q_3, q_4\}$
$\{q_3, q_4\}$	$\{q_4\}$	$\{q_3, q_4\}$
Ø	Ø	Ø

How many of these rows should be marked as accepting states?

Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Numerical Type | Award: 2 | Penalty: 0 | Theory of Computation



Suppose we run the above NFA on the string 10110. How many of the following statements are true?

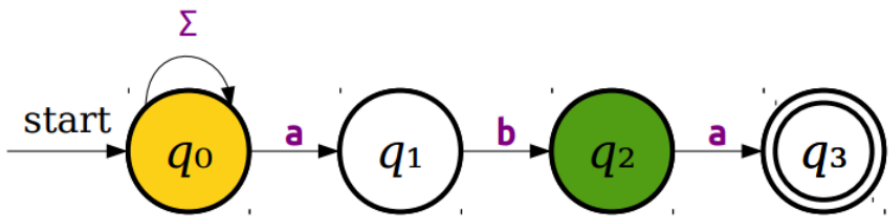
- There is at least one computation that finishes in an accepting state.
- There is at least one computation that finishes in a rejecting state.
- There is at least one computation that dies.
- This NFA accepts 10110.

- This NFA rejects 10110.

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

We transform the following NFA into an equivalent DFA using the subset construction (or the powerset construction) algorithm.



	a	b
$\{q_0\}$	$\{q_0, q_1\}$	$\{q_0\}$
$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_2\}$
$\{q_0, q_2\}$		

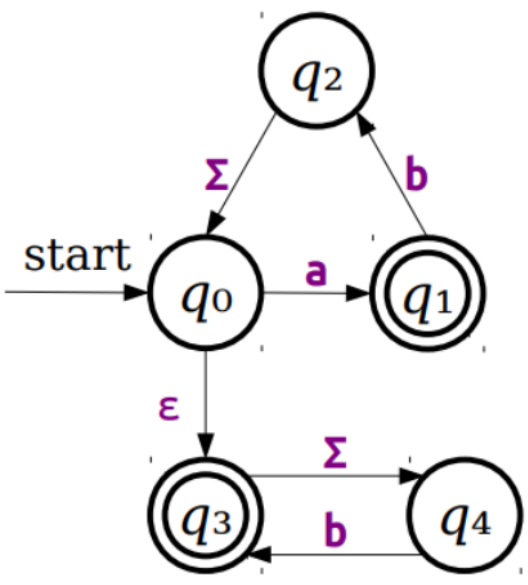
What should this row look like?

A. $\{q_0, q_2\}, \{q_0, q_3\}, \{q_0\}$
B. $\{q_0, q_2\}, \{q_0, q_3\}, \{q_0, q_2\}$
C. $\{q_0, q_2\}, \{q_0, q_1, q_3\}, \{q_0\}$
D. $\{q_0, q_2\}, \{q_0, q_1, q_3\}, \{q_0, q_2\}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

We transform the following NFA into an equivalent DFA using the subset construction (or the powerset construction) algorithm.



	a	b
$\{q_0, q_3\}$	$\{q_1, q_4\}$	$\{q_4\}$
$\{q_1, q_4\}$	Ø	$\{q_2, q_3\}$
$\{q_4\}$	Ø	$\{q_3\}$
$\{q_2, q_3\}$		

What should this row look like?

A. $\{q_2, q_3\}, \{q_0, q_3, q_4\}, \{q_0, q_3, q_4\}$
B. $\{q_2, q_3\}, \{q_3, q_4\}, \{q_3, q_4\}$

C. $\{q_2, q_3\}, \{q_0, q_4\}, \{q_0, q_4\}$
D. $\{q_2, q_3\}, \{\varnothing\}, \{\varnothing\}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Numerical Type | Award: 2 | Penalty: 0 | Theory of Computation

Here's the operator precedence for regular expressions, from highest to lowest:

$$(\mathrm{R})$$
$$\mathrm{R}^*$$
$$\mathrm{R}_1\mathrm{R}_2$$
$$\mathrm{R}_1 \cup \mathrm{R}_2$$

Consider the regular expression

$$ab^*c \cup d$$

How many of the strings below are in the language described by this regular expression?

- $ababc$
- abd
- ac
- $abcd$

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 01sec | Discuss

Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

- Let $\Sigma = \{a, b\}$.
- Let $\mathrm{L} = \{w \in \Sigma^* \mid w$ contains at most one $a\}$.

Which of the following is a regular expression for L?

A. $b^*(a^* \cup \varepsilon)\, b^*$
B. $b^*ab^* \cup b^*$
C. $b^*(a \cup \varepsilon)\, b^*$
D. $b^*a^*b^* \cup b^*$

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Consider the following language E :

$$E = \{a^n \mathbf{b}^n \mid n \in \mathbb{N}\}$$

Which of the following is a regular expression for the language E defined above?

A. $a * b*$
B. $(ab)*$
C. $\varepsilon \cup ab \cup a^2b^2 \cup a^3b^3$
D. None of the above

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Numerical Type | Award: 2 | Penalty: 0 | Theory of Computation

For the following language over the indicated alphabet, construct a DFA with the fewest number of states that accepts precisely the strings that are in the indicated language. For the alphabet $\Sigma = \{a, b, c\}$, construct a DFA for the language $\{w \in \Sigma^* \mid w$ contains exactly two $c\}$ What is the number of states in the DFA?

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Numerical Type | Award: 2 | Penalty: 0 | Theory of Computation

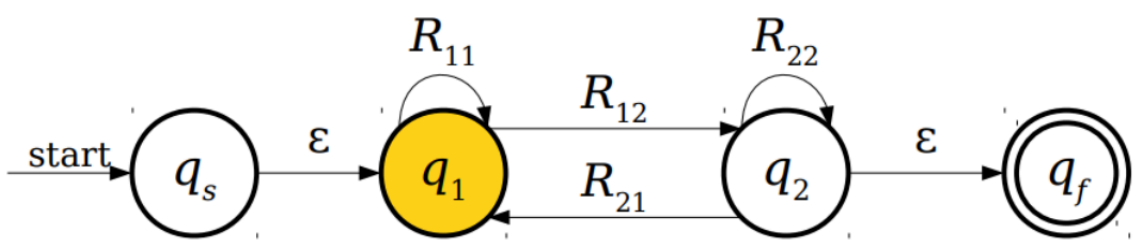
For the following language over the indicated alphabet, construct a DFA with the fewest number of states that accepts precisely the strings that are in the indicated language. For the alphabet $\Sigma = \{a, b, c, \ldots, z\}$, construct a DFA for the language $\{w \in \Sigma^* \mid w$ contains the word "cocoa" as a substring $\}$. What is the number of states in the DFA?

Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 00sec | Discuss

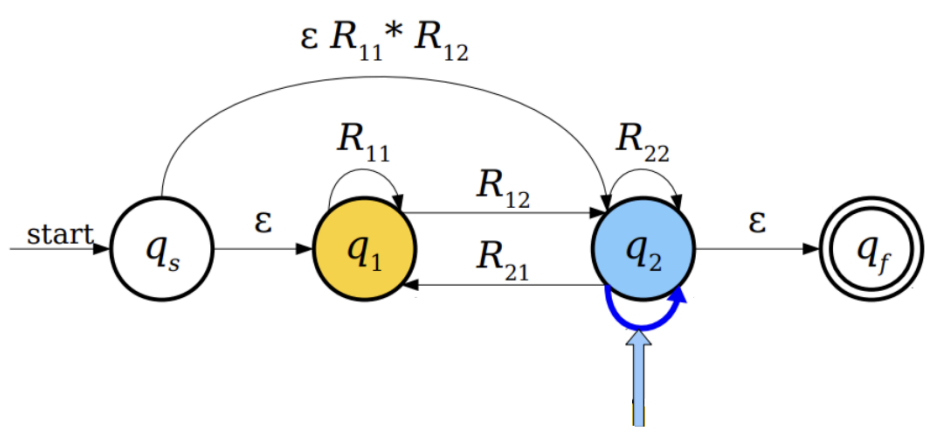
Q #14 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

The state elimination algorithm gives a way to transform a finite automaton (DFA or NFA) into a regular expression. It's a really beautiful algorithm once you get the hang of it, so we thought that we'd let you try it out on a particular example.

In the following Generalized NFA, we want to remove state q_1 :



If we remove state q_1, then we get two transitions(edges), one from q_s to q_1 (this new edge is shown in the figure below) and another from q_2 to q_2.



What regex should go on this edge?

A. $R_{12}R_{21}$

B. $R_{12}R_{22} * R_{21}$

C. $R_{21}R_{12}$

D. $R_{21}R_{11} * R_{12}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the following statements is/are true?

A. If L is a nonregular language, then there exists a nondeterministic finite automaton that accepts L.
B. If L_1 and L_2 are languages such that $\mathrm{L}_1 \subset \mathrm{L}_2$ and L_2 is nonregular, then L_1 must be nonregular.
C. The regular expressions $(\mathbf{a}+\mathbf{b})^*$ and $(\mathbf{a}^*\mathbf{b}^*)$ generate the same language.
D. All nonregular languages are infinite.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

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